

Robust Quadratic Formula for SOC Step-Length

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SOC Step-Length Problem

Given a current iterate $x = (x_0, x_1) \in \mathcal{Q}^n$ and a search direction $dx = (dx_0, dx_1)$, compute the largest step that keeps the iterate feasible

$$\alpha^* = \max\{\alpha \geq 0 : x + \alpha dx \in \mathcal{Q}^n\}$$

where the **second-order cone** is

$$\mathcal{Q}^n = \{(x_0, x_1) \in \mathbb{R} \times \mathbb{R}^{n-1} : x_0 \geq \|x_1\|\}$$

- x is always in the interior of $\mathcal{Q}^n$ during an interior-point method
- α^* bounds the step so that $x + \alpha^* dx$ stays inside the cone

Reducing to the Quadratic Formula

The membership condition for $x + \alpha dx \in \mathcal{Q}^n$ is

$$(x_0 + \alpha dx_0)^2 \geq \|x_1 + \alpha dx_1\|^2$$

Expanding and collecting by powers of α :

$$\underbrace{(dx_0^2 - \|dx_1\|^2)}_a \alpha^2 + \underbrace{2(x_0 dx_0 - x_1^\top dx_1)}_b \alpha + \underbrace{(x_0^2 - \|x_1\|^2)}_c \geq 0$$

Since $x \in \mathcal{Q}^n$, we have $c = x_0^2 - \|x_1\|^2 \geq 0$, so $\alpha = 0$ is always feasible. The maximum feasible α is the **smallest positive real root** of

$$a\alpha^2 + b\alpha + c = 0$$

which is found via the **quadratic formula**: $\alpha = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

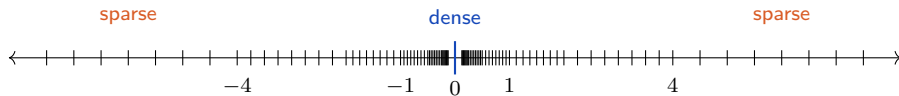
Floating-Point Error Scales with Magnitude

Floating-point stores a fixed number of **significant bits**, so the absolute spacing between consecutive numbers scales with magnitude

IEEE 754 guarantees relative error $\leq \epsilon_{\text{mach}}$ per operation, where $\hat{x}$ is the **rounded floating-point result** of computing x :

$$\hat{x} = x(1 + \epsilon), \quad |\epsilon| \leq \epsilon_{\text{mach}}$$

So the **absolute error** scales with $|x|$: $|\hat{x} - x| \leq \epsilon_{\text{mach}} \cdot |x|$



Each tick is a representable number (3-bit mantissa, for illustration)

Catastrophic Cancellation

To find the smaller root when $b > 0$, the standard formula gives

$$\alpha^* = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Problem: as the IPM converges, the iterate approaches the cone boundary, so

$$c = x_0^2 - \|x_1\|^2 \longrightarrow 0 \implies \sqrt{b^2 - 4ac} \approx |b|$$

The numerator $-b + \sqrt{b^2 - 4ac}$ subtracts two **nearly equal floating-point numbers**, causing **catastrophic cancellation** where the significant digits cancel and only noise remains

- Relative error in α^* can be enormous even though $|a|$, $|b|$, $|c|$ are well-represented
- This happens precisely where accuracy matters most: near the optimal solution

Why Catastrophic Cancellation Occurs

In floating point, $\sqrt{b^2 - 4ac}$ is computed with rounding error. Floating-point gives relative error, so the absolute error scales with the magnitude of the result:

$$\hat{S} = \sqrt{b^2 - 4ac} + \delta, \quad |\delta| \lesssim \epsilon_{\text{mach}} \cdot |b|$$

where the bound uses $\sqrt{b^2 - 4ac} \approx |b|$ when $c \approx 0$. The computed numerator is therefore

$$-b + \hat{S} = \underbrace{(-b + \sqrt{b^2 - 4ac})}_{\text{true result}} + \delta$$

As $c \rightarrow 0$ the true result approaches toward zero while $|\delta| \lesssim \epsilon_{\text{mach}} \cdot |b|$ stays fixed, so the relative error diverges:

$$\frac{|\delta|}{-b + \sqrt{b^2 - 4ac}} \lesssim \frac{\epsilon_{\text{mach}} \cdot |b|}{-b + \sqrt{b^2 - 4ac}} \rightarrow \infty$$

Once the true result is smaller than the roundoff in $\hat{S}$, the computed numerator is **pure noise** and no significant digits remain

The Robust Quadratic Formula (Citardauq)

Rewrite to ensure each root is computed by dividing two quantities of the **same sign**

Step 1. Compute $\sqrt{d} = \sqrt{b^2 - 4ac}$ and define

$$t = \begin{cases} -b - \sqrt{d} & \text{if } b \geq 0 \\ -b + \sqrt{d} & \text{if } b < 0 \end{cases}$$

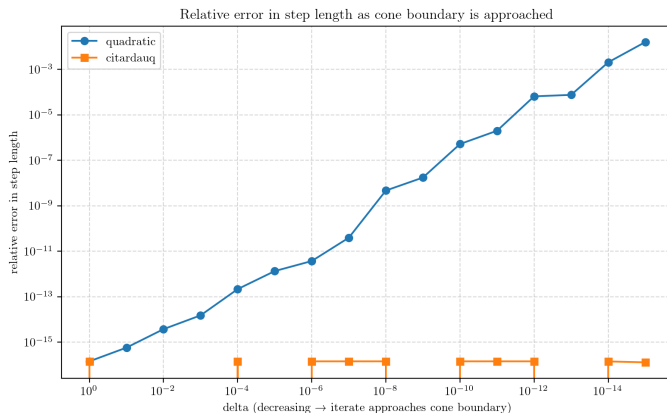
By construction, t and b always have opposite signs, so t is computed **without cancellation**.

Step 2. Recover both roots via Vieta's formulas ($r_1 r_2 = c/a$):

$$r_1 = \frac{2c}{t}, \quad r_2 = \frac{t}{2a}$$

Negative roots are discarded and $\alpha^* = \min(r_1, r_2)$ over positive roots.

Error Comparison



Standard quadratic formula incurs large error as iterate approaches cone boundary, but citardauq formula remains accurate throughout